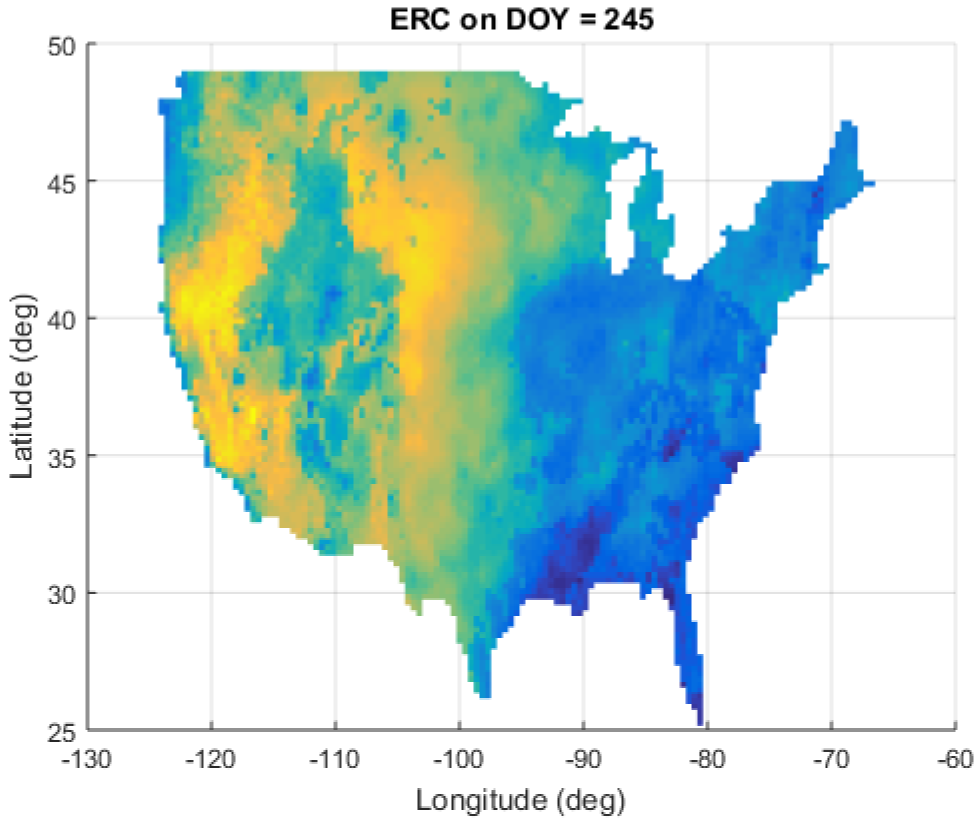
```
% Load spatial meshes
load('C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Omega/OUTPUT/
meshes.mat'); %Consult structure in workspace for formatting
fnam = 'erc_2012.nc'; %ERC file
if ~exist('erc_2012','var') %Save time if variable is already in
 workspace
    disp(['Loading  ' fnam ' into workspace...']); %Report to user
    erc_2012 = ncread(fnam,'energy_release_component-g'); %Read file
end
if ~exist('lat','var') || ~exist('lon','var') %Skip to save time
    lat = ncread(fnam,'lat'); %Load latitude
    lon = ncread(fnam,'lon'); %Load longitude
end
if ~exist('day','var') %Skip to save time
    day = ncread(fnam,'day'); %Load day of year (DOY)
end
offset = datenum(2012,1,1); %Offset for this year
% Build time mesh with even spacing
T0 = datenum(2012,8,25)-offset; TF = datenum(2012,9,2)-offset; TSTEP =
 0.25;
% MATLAB style grid for interpolation and plotting
[lonMesh,latMesh] = meshgrid(lon,lat);
% MATLAB style grid as target mesh
[ERC(13).xq,ERC(13).yq] =
 meshgrid(unique(mesh(13).grid(:,1)),unique(mesh(13).grid(:,2)));

% INTERPOLATE!
disp(['Interpolating ' mesh(13).name ' in space...']); %Report to user
nn = 0; %Initialize time index
for tt = T0:TF %For each time in mesh
    nn = nn + 1; %Iterate time index
    disp(['- Working on DOY = ' num2str(tt)]); %Report to user
    % Scattered interpolant to our mesh
    F =
 scatteredInterpolant(reshape(lonMesh,numel(lonMesh),1),reshape(latMesh,numel(latM
    ERC(13).zzCoarse(:,:,nn) = F(ERC(13).xq,ERC(13).yq); %Evaluate
 interpolation function at desired nodes
    % Plotting

  %surf(ERC(13).xq,ERC(13).yq,ERC(13).zzCoarse(:,:,nn),'EdgeColor','none');
 view(0,90);
    %title(['DOY = ' num2str(tt)]); pause(1.5);
end

% REFINE TIMESCALE!
disp(['Refining ' mesh(13).name ' in time...']); %Report to user
ERC(13).tq = (T0:TSTEP:TF)'; %Refine time interval
% Initialize space for ERC values on refined grid
ERC(13).zzFine = NaN*ones(mesh(13).nx,mesh(13).ny,numel(ERC(13).tq));
% Loop through nodes
for xx = 1:mesh(13).nx %For each horizontal coordinate
    for yy = 1:mesh(13).ny %For each vertical coordinate
```

```
        vecA = ERC(13).zzCoarse(xx,yy,:); %Parse coarse values
        vecA = reshape(vecA,numel(vecA),1); %Flatten to vector
        if sum(~isnan(vecA)) > 1 %If there is more than one data point
            vecB = spline((T0:TF)',vecA,ERC(13).tq); %Use splines to
 interpolate across time
            ERC(13).zzFine(xx,yy,:) = vecB; %This is the new ERC time
 series for point xx,yy
        end
    end
end

% Initialize movie capture
writerObj = VideoWriter('C:\Users\Alex\Documents\MATLAB\ALOC8\Datasets
\Gridded ERC\OUTPUT\vid.avi');
open(writerObj);

% EXTRAPOLATE TO OUTER BOUNDARY
disp(['Extrapolating ' mesh(13).name ' in space to OB...']); %Report
 to user
nn = 0; %Initialize time index
fig = gcf(); %Get current figure
% Initialize movies
movie(numel(T0:TSTEP:TF)) = struct('cdata',[],'colormap',[]);
for tt = T0:TSTEP:TF %For each refined time step
    nn = nn + 1; %Interate time index
    F =
 scatteredInterpolant(reshape(ERC(13).xq,numel(ERC(13).xq),1),reshape(ERC(13).yq,n
    ERC(13).zzOB(:,:,nn) = F(mesh(13).OB(:,1),mesh(13).OB(:,2));
    scatter3(mesh(13).OB(:,1),mesh(13).OB(:,2),ERC(13).zzOB(:,:,nn));
 view(0,90);

 surf(ERC(13).xq,ERC(13).yq,ERC(13).zzFine(:,:,nn),'EdgeColor','none');
 view(0,90);
    xlabel('Longitude (deg)'); ylabel('Latitude (deg)');
    title(['ERC on DOY = ' num2str(tt)]); pause(0.5);
    movie(nn) = getframe(fig);
    writeVideo(writerObj,movie(nn));
end
close(writerObj);
% Save output as MATLAB workspace variable
save('C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Gridded ERC/
OUTPUT/ERC.mat','ERC');
```

Interpolating Continental US in space...
- Working on DOY = 237
- Working on DOY = 238
- Working on DOY = 239
- Working on DOY = 240
- Working on DOY = 241
- Working on DOY = 242
- Working on DOY = 243
- Working on DOY = 244
- Working on DOY = 245
Refining Continental US in time...

```
Extrapolating Continental US in space to OB...
```



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